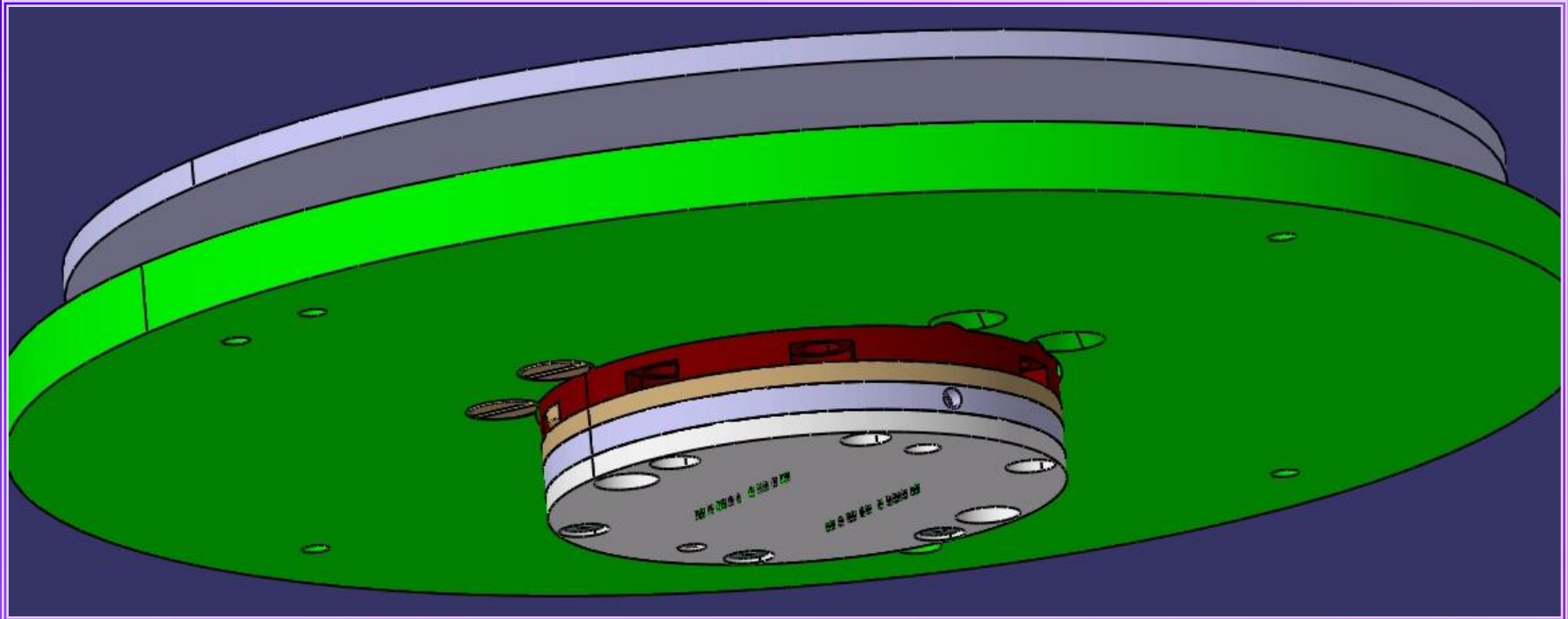


Vertical Probing Technology Trio™ Probe Card

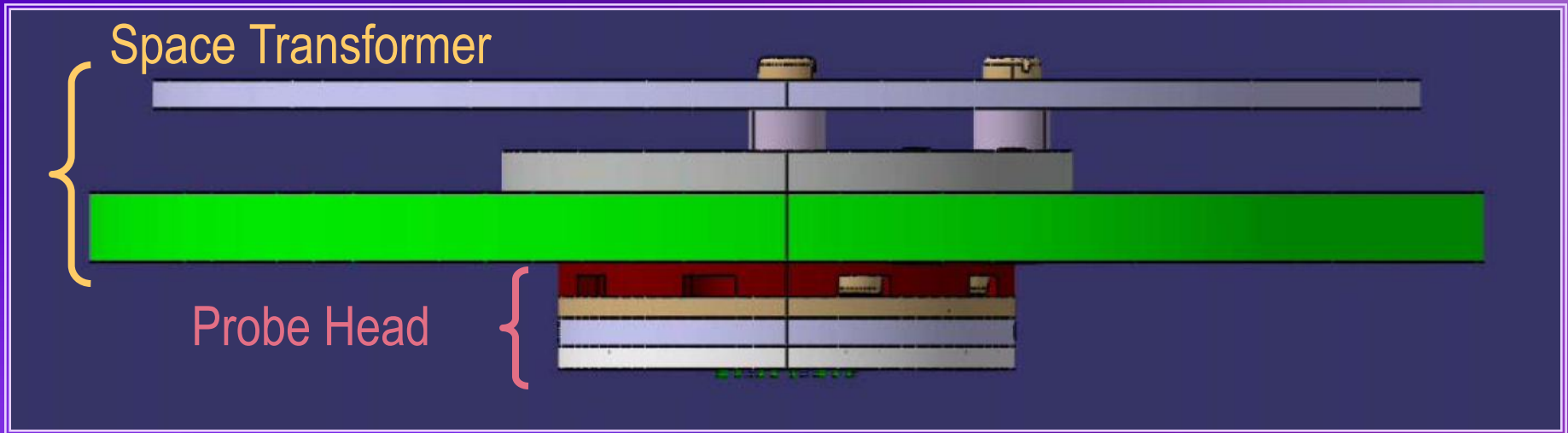


What is a Vertical Probe Card?

- ❑ Any medium that translates electrical paths from Die to PCB (Tester Channel).
- ❑ Ability to Probe Area Array, Flip Chip, Solder Bumps
- ❑ Ability to Probe Multiple Devices in parallel



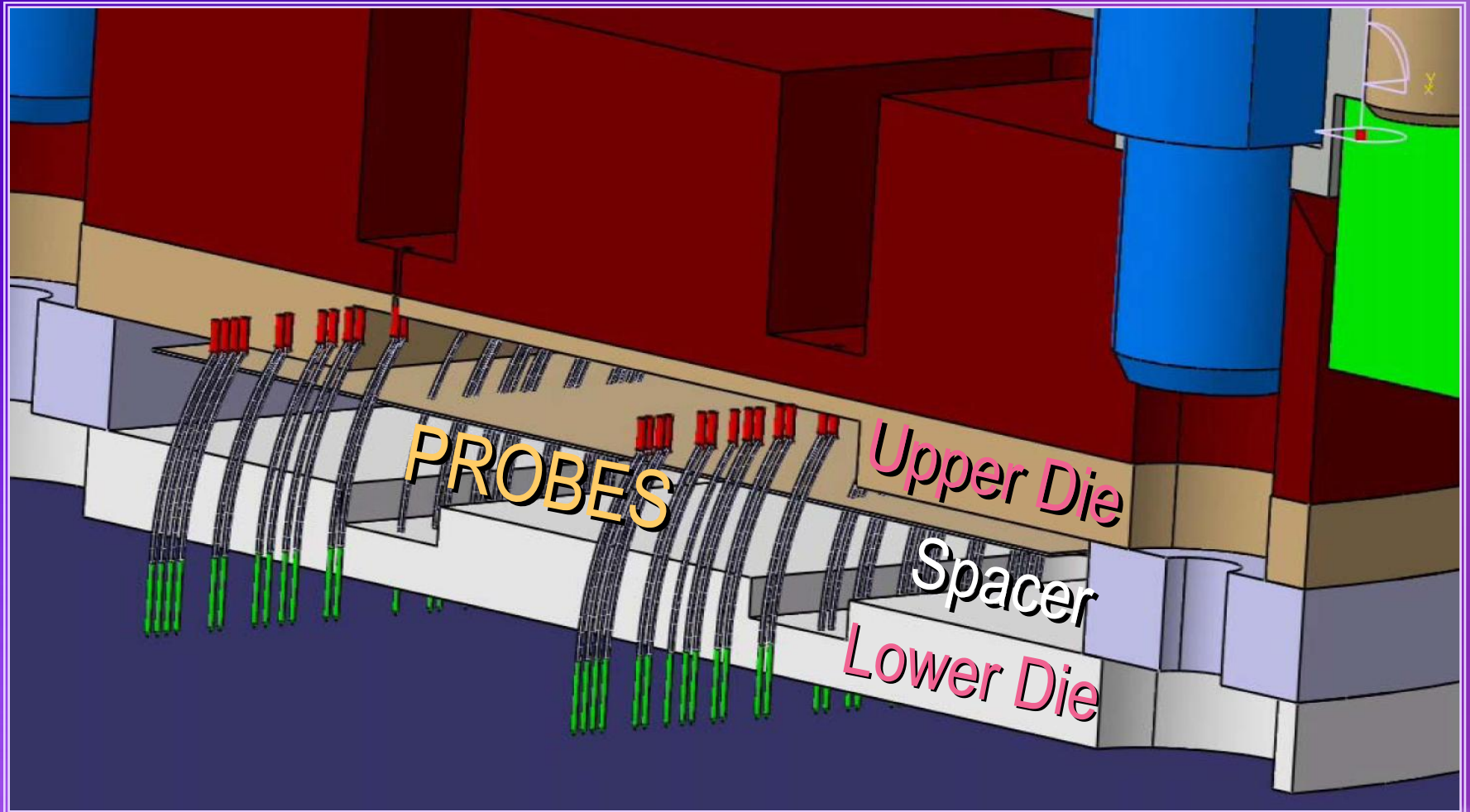
Probe Card Assembly



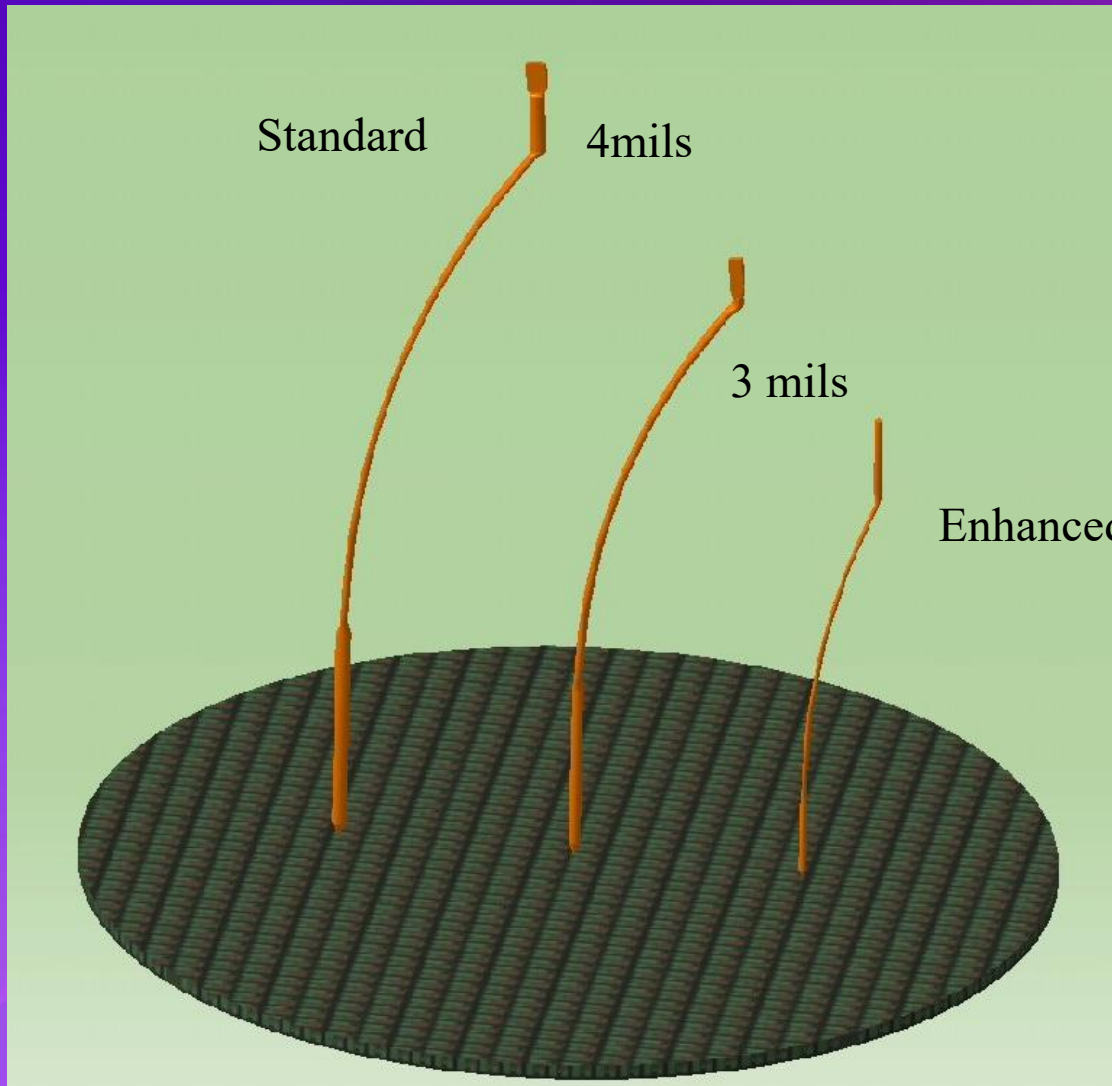
Types of Space Transformers used:

- Wired
- Rigid (SV Substrate/ CS Substrate)
- Direct attach/ Modular

Vertical Probe Head



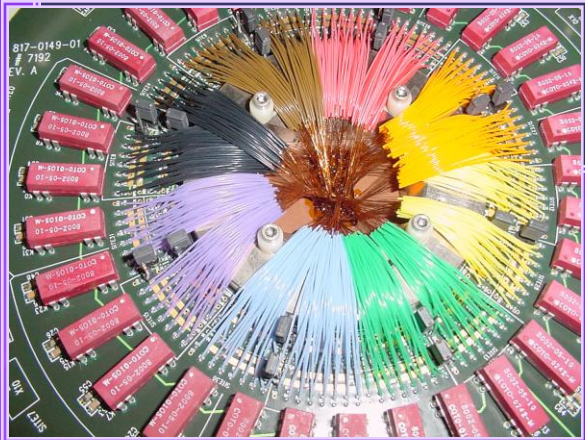
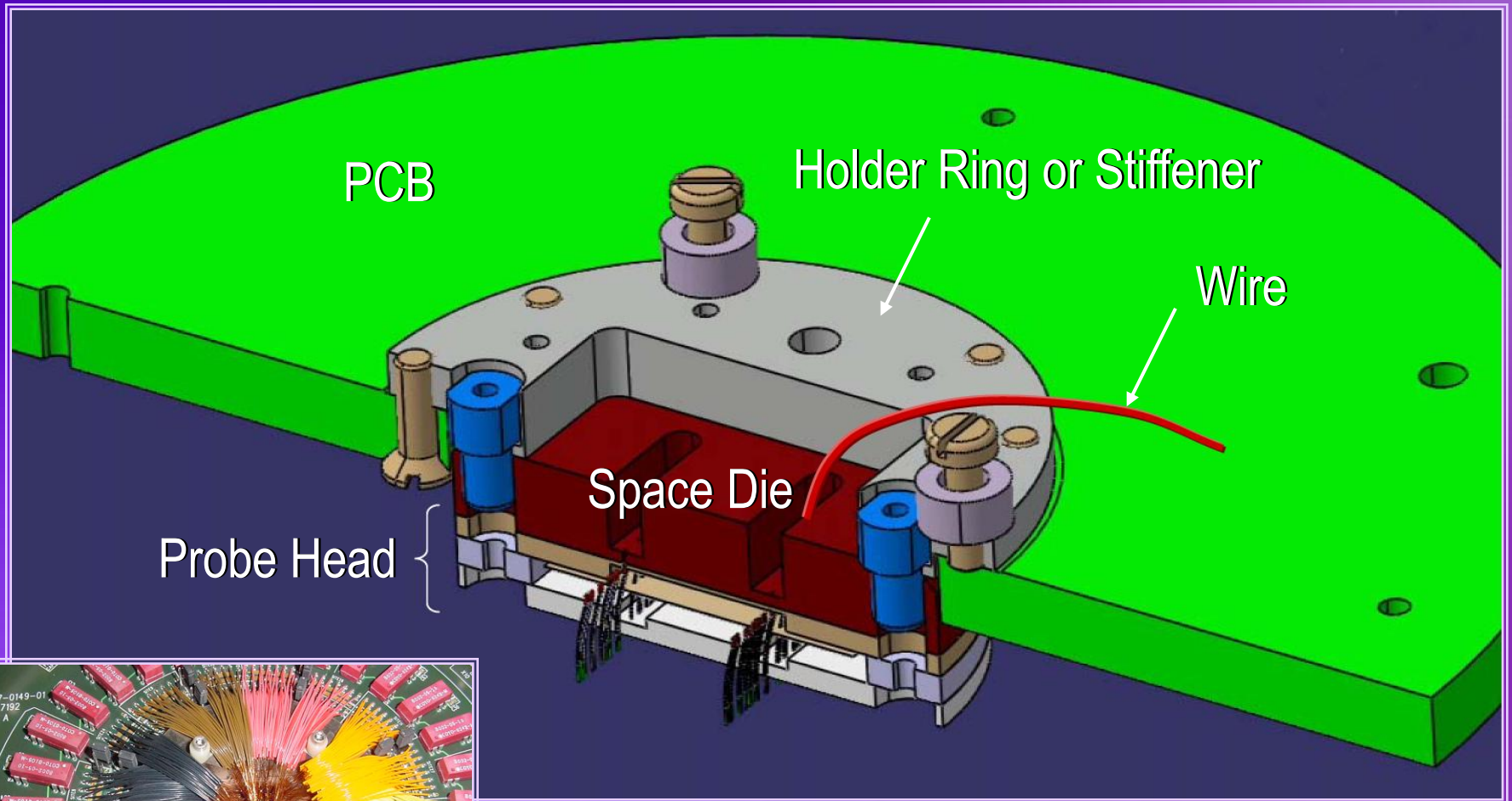
Probe Wires



PROBE TYPE	
☐ 4 MIL	☒ POINTED
☐ 3 MIL	☐ FLAT
☒ 2 MIL	☐ Paliney 7
	☒ PowerPlus™



Hand Wired Vertical Card



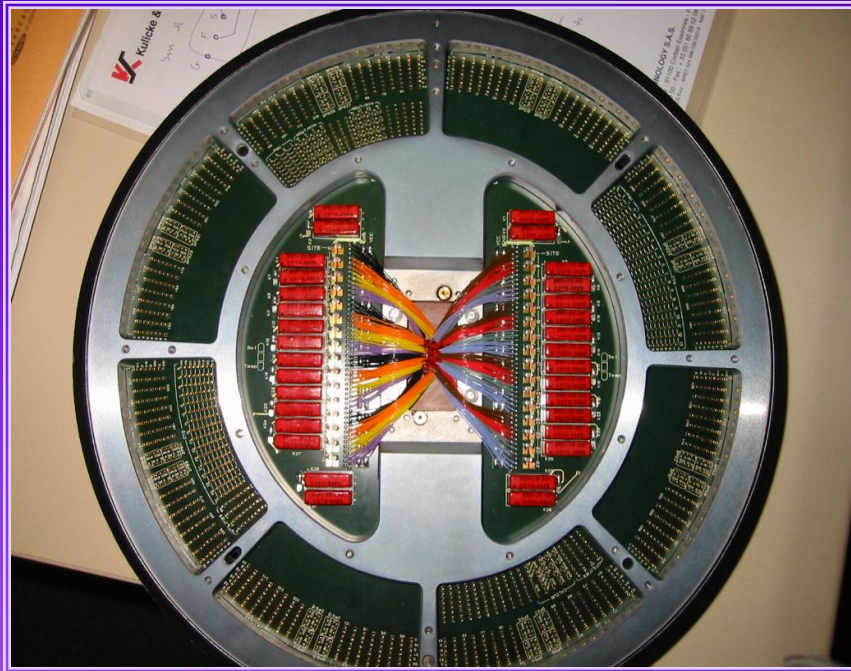
Wired picture

Hand Wired Vertical Card

- ❑ PCB must be countered in center
- ❑ Have Space Die
- ❑ Use Holder Ring/ Stiffener to hold the PH and PCB
- ❑ Use wires to make electrical connection from PH with the PCB
- ❑ Must have Solder Pads and CH



Example of Complex Stiffener Design: Combo Stiffener + Space Die Holder



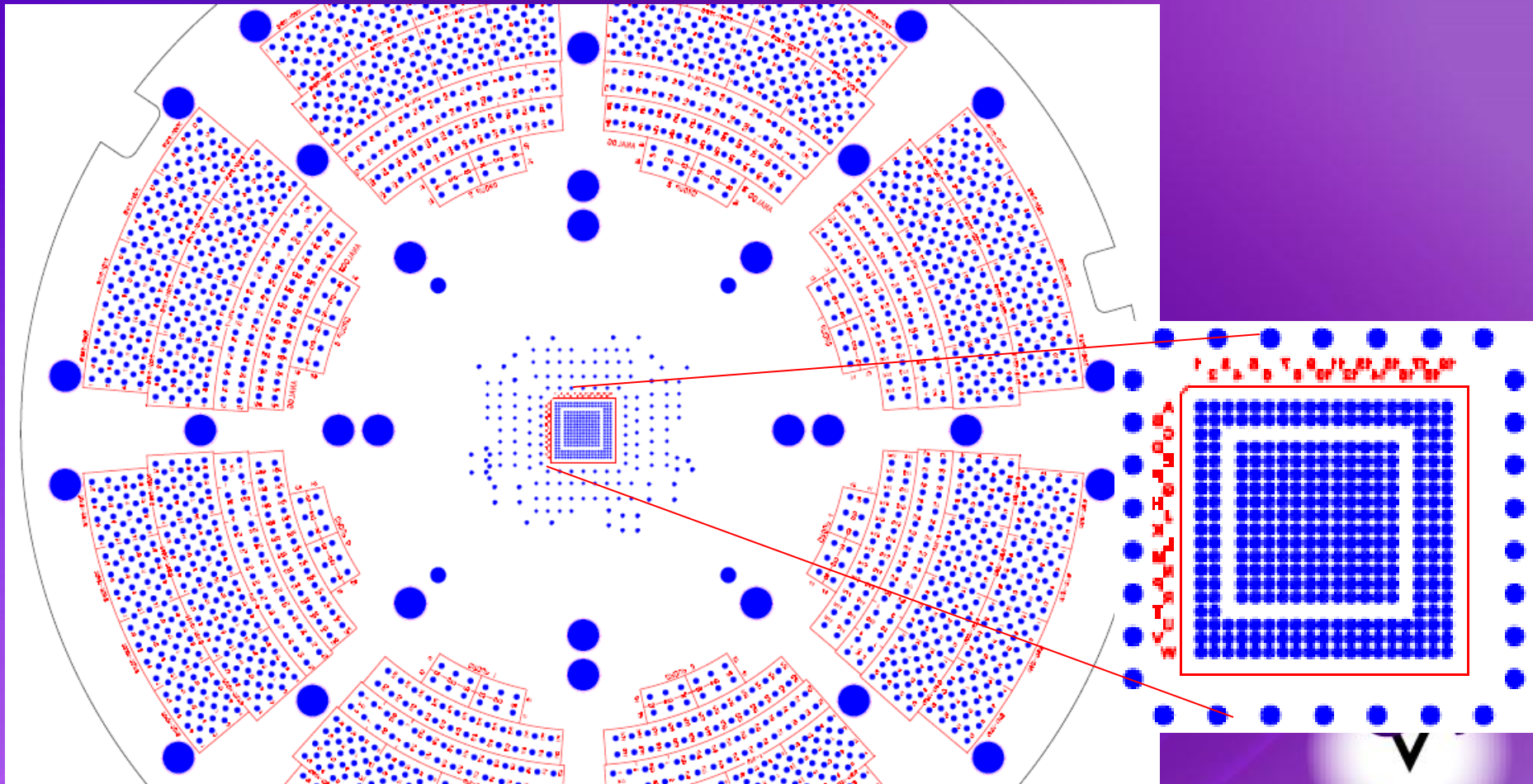
Wired Space Transformers Summary

- ❑ Uses Wire to Translate the Electrical Path from the Head to the PCB
- ❑ Works Best for Lower Point Counts <2000
- ❑ Can use Generic PCBs
- ❑ Lower Costs
- ❑ Field Replaceable Heads
- ❑ SV Factory Replaceable Probes

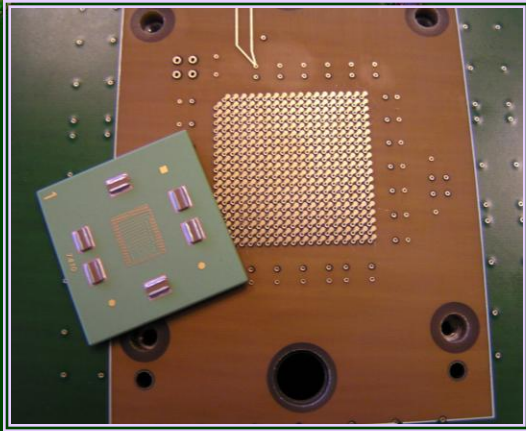


Rigid (SV Substrate/ CS Substrate)

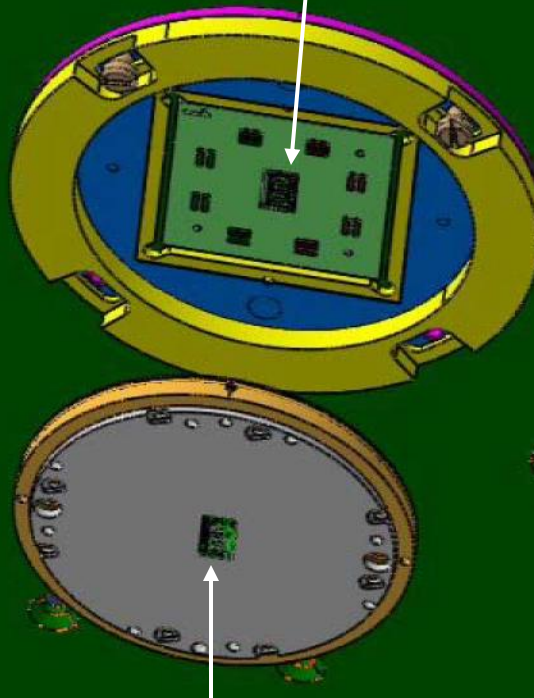
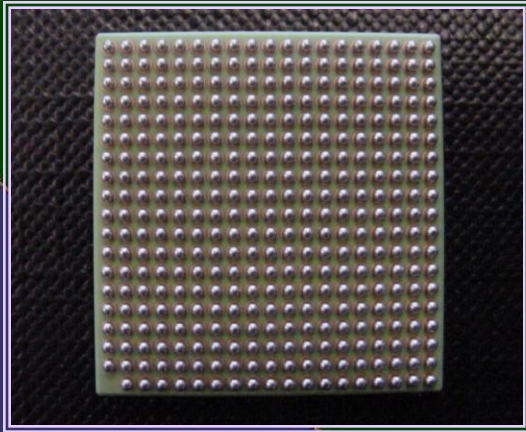
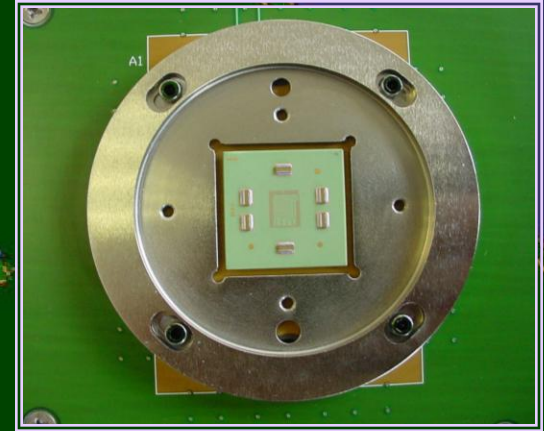
- PCB don't have to counter bore



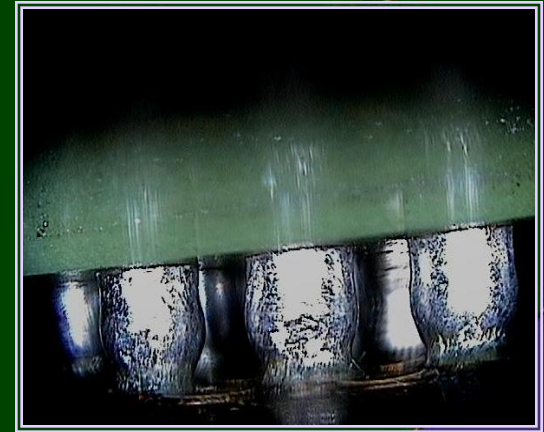
Rigid (SV Substrate/ CS Substrate)



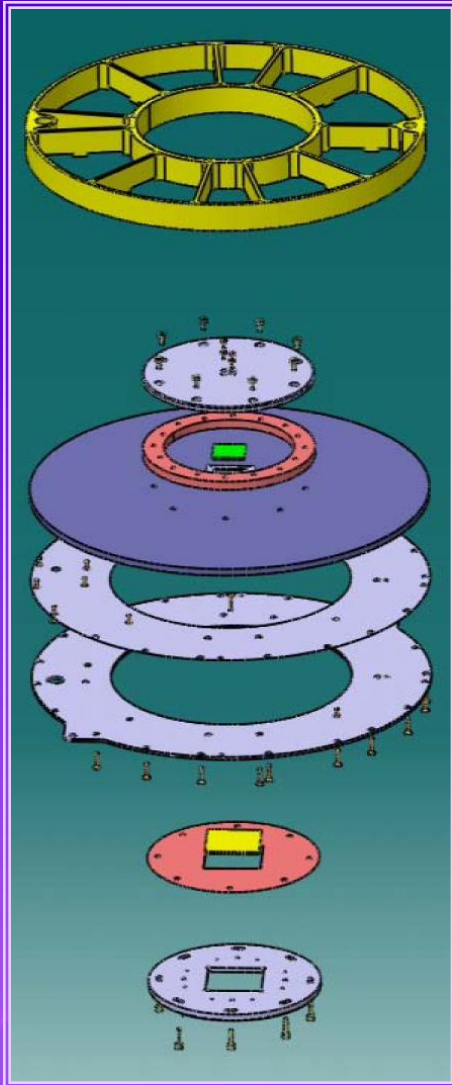
Substrate (MLC,
MLO, LTCC...)



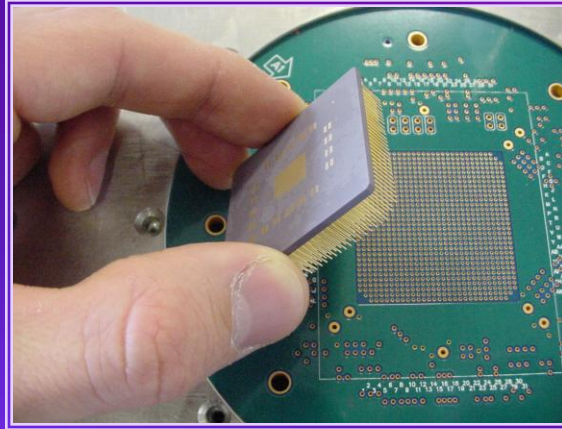
Probe Head



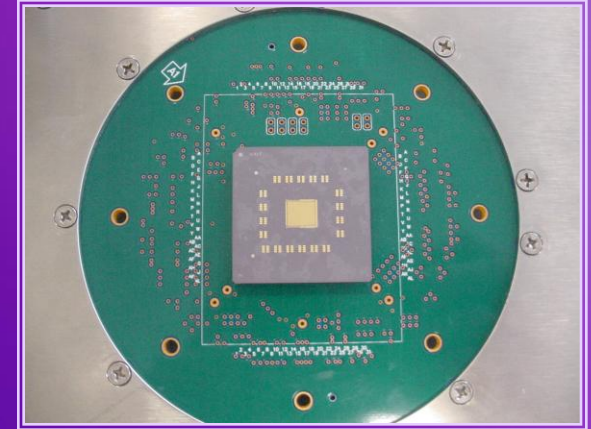
Rigid (SV Substrate/ CS Substrate)



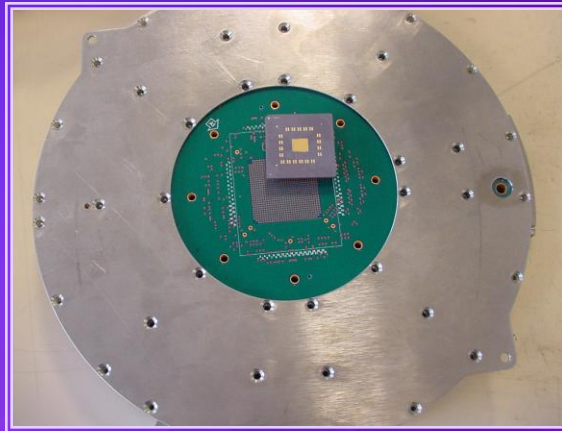
1



3



2



4



Rigid (SV Substrate/ CS Substrate)

Summary

- ❑ Uses a Rigid Substrate to Translate the Electrical Path from the Head to the PCB
- ❑ Better electrical performance
- ❑ Higher point counts >1000
- ❑ Must have custom PCB
- ❑ Probe depth is fixed by the thickness of the assembly
- ❑ Long design cycle
- ❑ Moderate to high cost
- ❑ Short electrical paths possible

